bloons

board

Cell
- position:Positon

+ Cell(Position)
+getPosition():Positon

Position
-x:int
-y:int

+Position(int,int)
+getX():int
+getY():int
+equals(o:Object):boolean

Board
heigth: int
width: int
Allballoons : List <Balloons>
grid : Cell[][]

+Board(int,nt)
+*calculPath():void*
+getHeigth():int
+getWidth():int
+*getPath()* : List <Position>
+getCell(p : Position) : Cell
+displayBoard()
+displayPath()
+getGridCells() : Cell[][]
+getAllballoons() : List<Balloons>
+generateAllballons(int)
+*createBalloonForPath(int, Level) : Balloon*

ClassicalBoard
path : List<Position>

+ClassicalBoard(int;int)
+calculPath():void
+getPath() : List <Position>
+isValid(Position) : boolean
+chooseValidNeighborPosition(Position) : Position
+createBalloonForPath(int, Level) : Balloon

RandomBoard
allPath : List<List<Position>>

+RamdomBoard(heigth:int,width:int)
+calculPath():void
+getPath() : List <Position>
+generatePath(int)
+getAllPaths() : List<List<Position>>
+getPathAt(int) : List<Position>
+displayBoard(List<Position>)
+createBalloonForPath(int, Level) : Balloon

balloon

Balloon
- path : List<Position>
- position : Position
- level : Level
- health : int
- speed : double
- currentCellIndex : int
- prgresse : doubel

+Balloon (Position, Level)
+ getLevel() : Level
+ getPosition : Position
+ getSpeed() : double
+ getHealth() : int
+ getProgresse() : double
+ getCurrentCellIndex()
+ setHeath()
+ isDestroyed() : boolean
+ setTrajectory(List<Position>)
+ checkNextCell()
- updateBalloonPosition()
+ moveBalloon()
+ checkEndOfPath() : boolean
+ randomChoiceSpeed() : double
+ getPath() : List <Position>

utils

Direction <enum>
+ DOWN
+ RIGTH
+ TOP
- dx : int
- dy : int

+ Direction(int, int)
+ generateNeighbor(Position): Position

Level <enum>
+ NIV1
+ NIV2
+ NIV3
- resistance : int

- Level(int)
+ getResistance() : int
+ generateLevel() : Level

Projectile<enum>
+DART
+POINTEDDART
+VERYSHARPDART
+NEEDLE
+BOMB
+EXTRABOMB
-niveauDegat:int

-Projectile(int)
+getNiveau():int

Effect<enum>
+ IMMEDIAT
+ DISTANCE

tower

projectileTower

SlowDownTower

+SlowDownTower(double,double,Effect,double,Position)
+attackBallon(Balloon)

Tower
#coast: double
#range :double
#effect:Effect
#position: Position
#cadence: double
#timeLastAttack:long
#balloons: List<Balloon>
#ballonInRange:List<Balloon>

+Tower(double,double,Effect,double, position)
+getEffect(): Effect
+getcadence():double
+ getRange(): double
+setCadence()
+getCost(): double
+getPosition():Position
+inRange(Balloon):boolean
+calculDistance(Balloon):
+addBalloon(<ListBalloon>)
+getAllBalloonInRange:
+getAllBalloonInRange:
+updateAttackTime(long)
+getTimeLastAttack():long
+canAttack(long)
getBalloonInRange()
+toString():String
+*attackBallon(Balloon long)*

IceTower
-reelSpeed:double

+IceTower(double,double,Effect,double,Position)
+ freeze(Balloon)
+attackBallon(Balloon,long)

DartMonkeyTower

+DarkMokey(double,double,Effect,double, Positon,Projectile)
+attackBalloon(Balloon,long):void

BombTower

+BombTower(double,double,Effect,double, Position,Projectile)
+attackBalloon(Balloon,long):void

SniperMonkeyTower

+SnipeMonkeyTower(double,double, Effect,double,Position,Projectile)
+InRange(Balloon)

ProjectileTower
+projectile:Projectile

+ProjectileTower(double,double,Effect,double, Position,Projectile
+attackBalloon(Ballon long):void
+getProjectile(Projectile):void
+setProjectile(Pojectile)

TackShooterTower

+TackShooterTower(double,double, Effect,double,Position,Projectile)

SuperMonkeyTower

+SuperMonkeyTower(double,double, Effect,double,Position,Projectile)